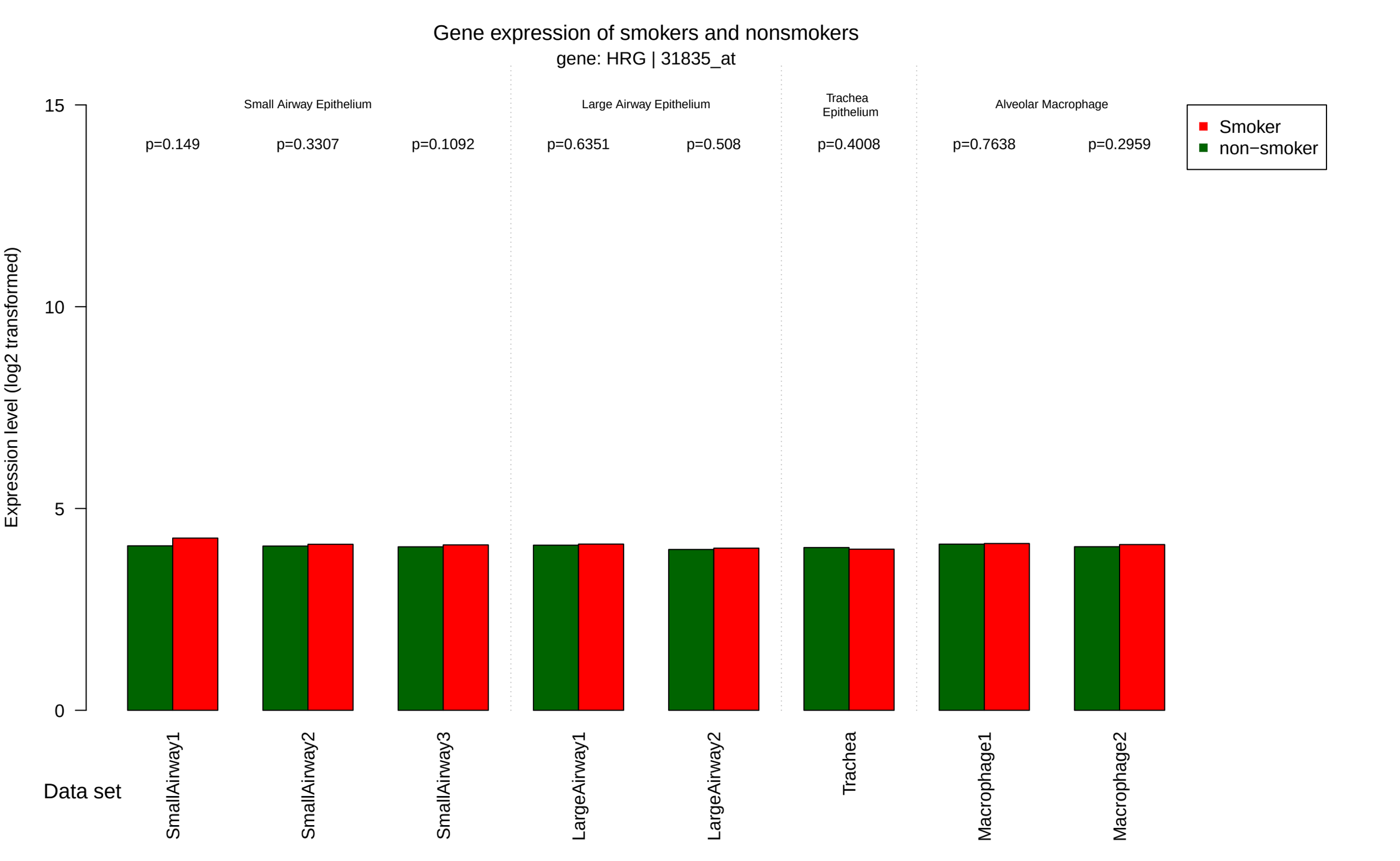
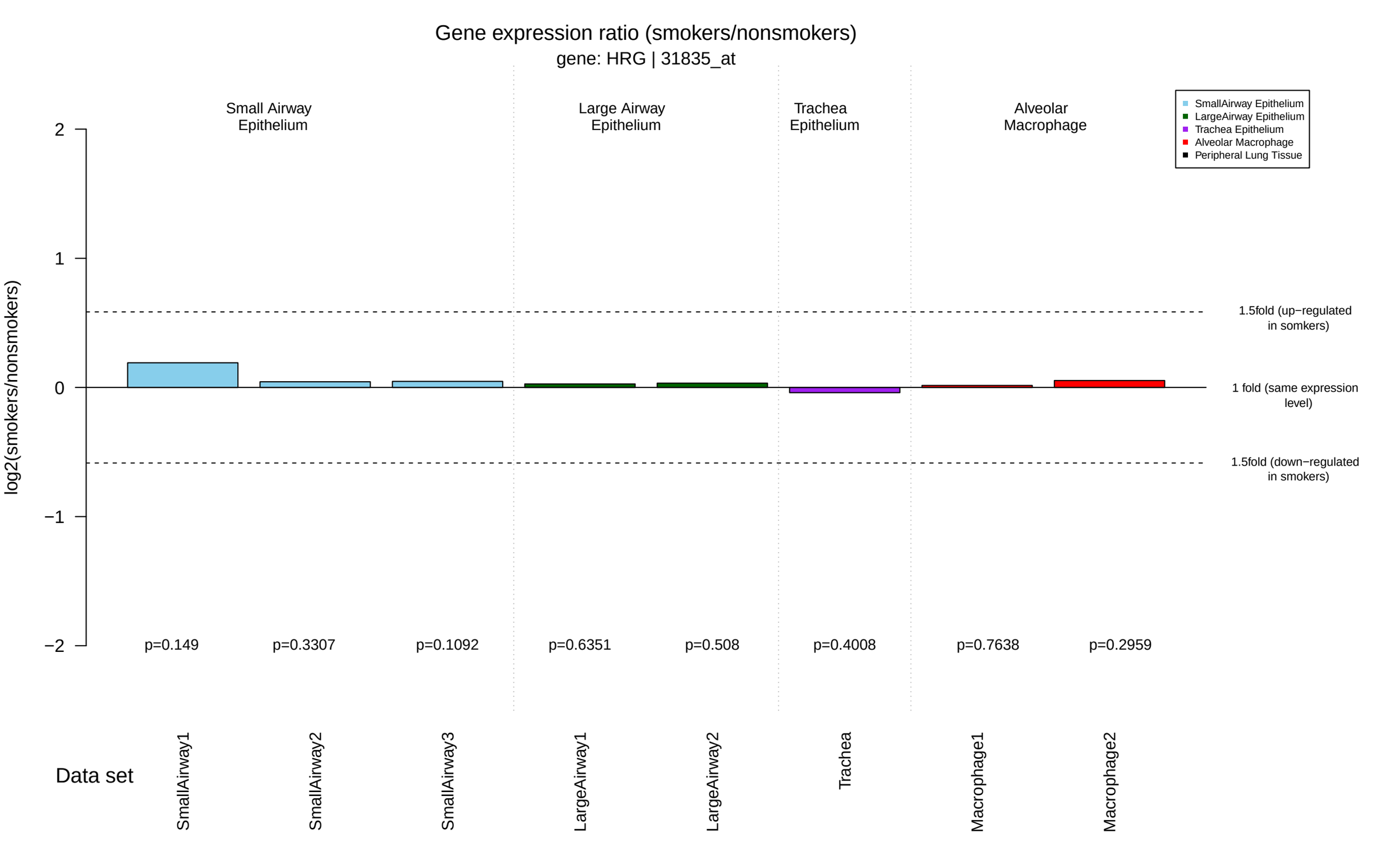
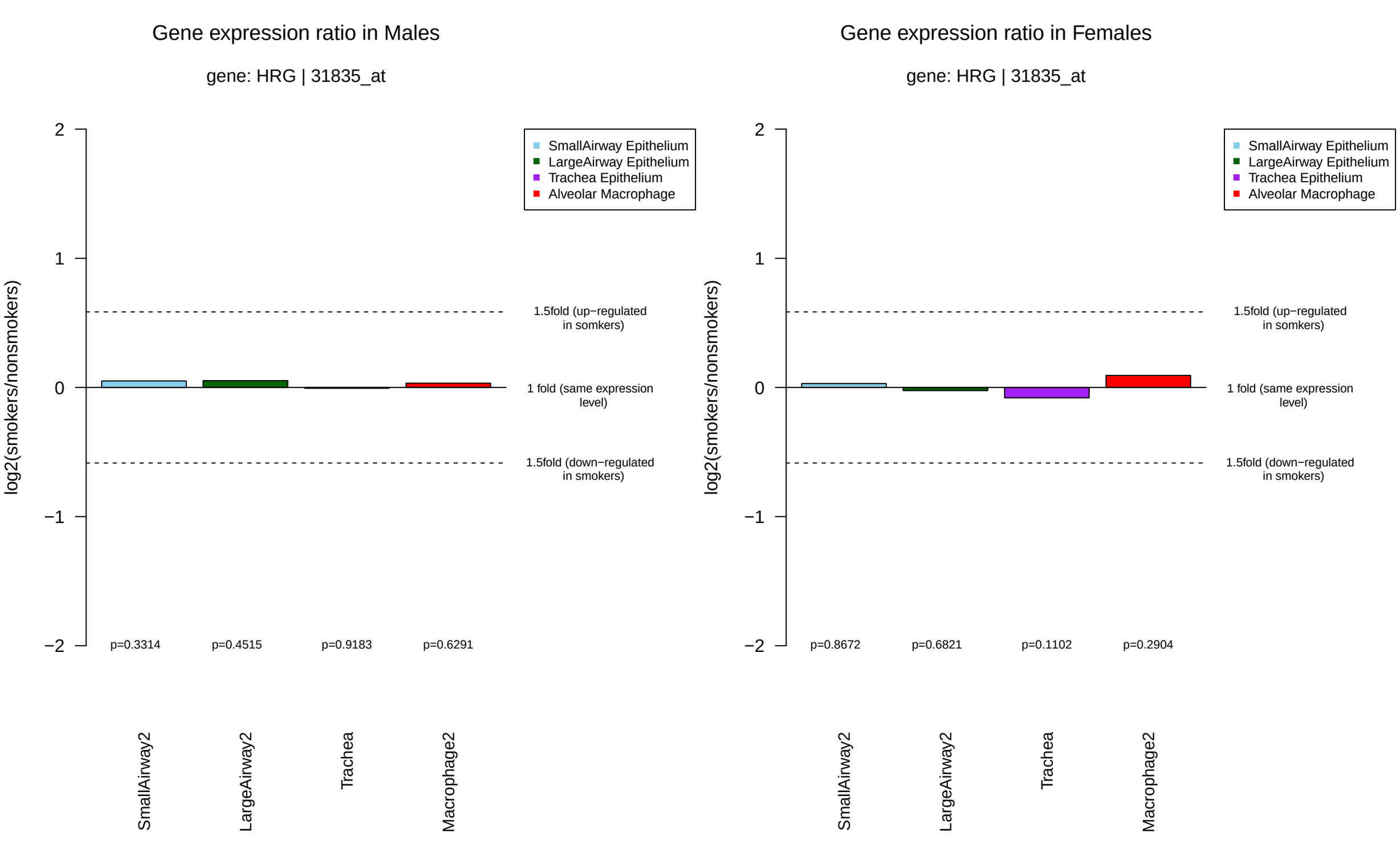
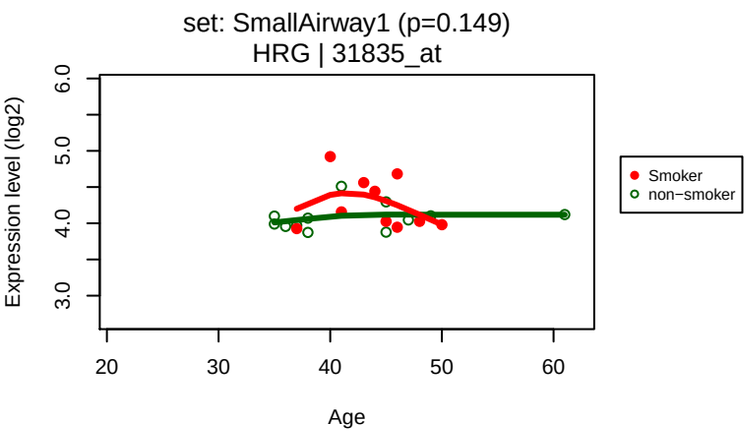
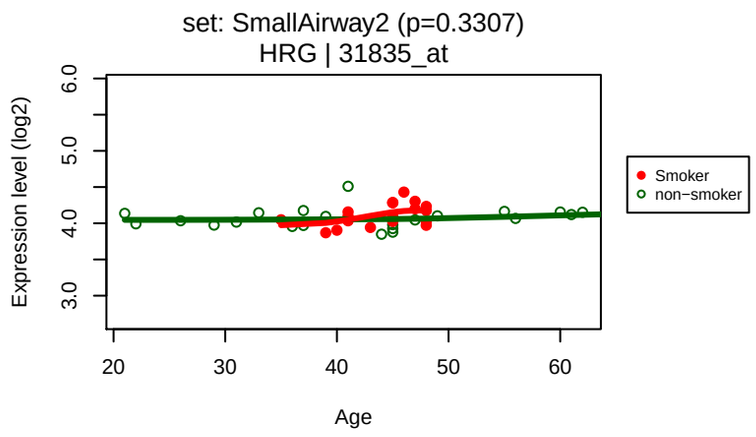
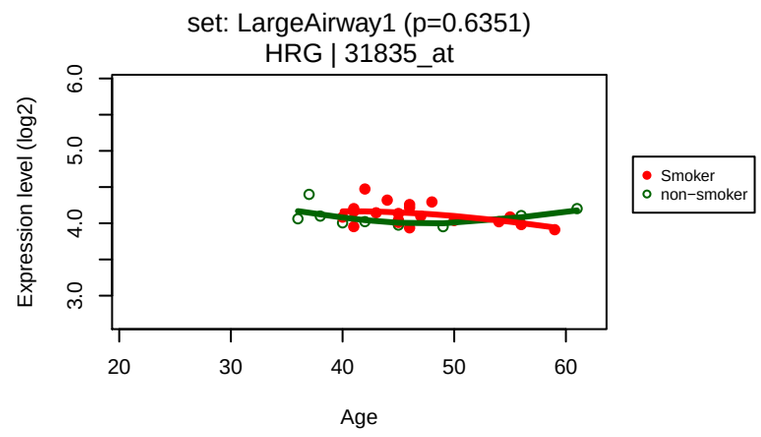

Gene expression of smokers and nonsmokers
gene: HRG | 31835_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.149
p=0.3307
p=0.1092
p=0.6351
p=0.508
p=0.4008
p=0.7638
p=0.2959
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

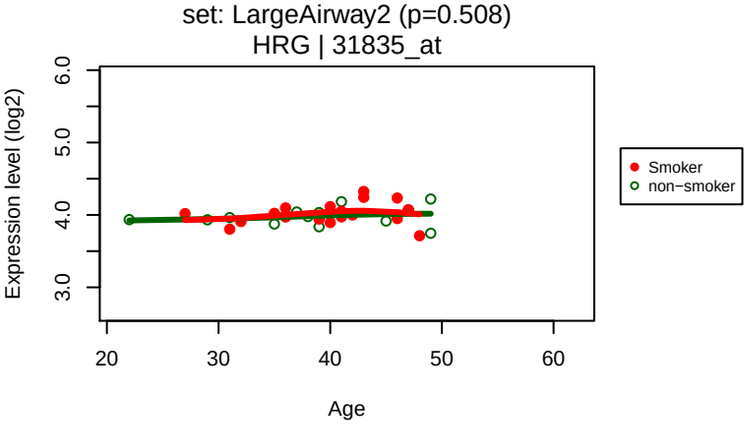

Gene expression ratio (smokers/nonsmokers)
gene: HRG | 31835_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.149
p=0.3307
p=0.1092
p=0.6351
p=0.508
p=0.4008
p=0.7638
p=0.2959
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

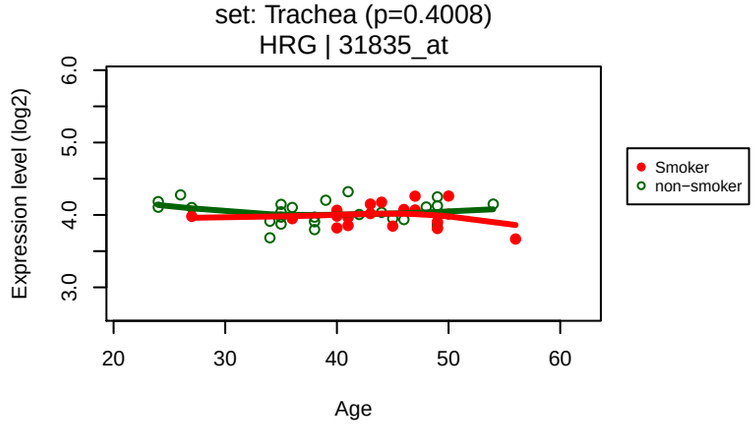

Gene expression ratio in Males
gene: HRG | 31835_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3314
p=0.4515
p=0.9183
p=0.6291
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HRG | 31835_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8672
p=0.6821
p=0.1102
p=0.2904
SmallAirway2
LargeAirway2
Trachea
Macrophage2

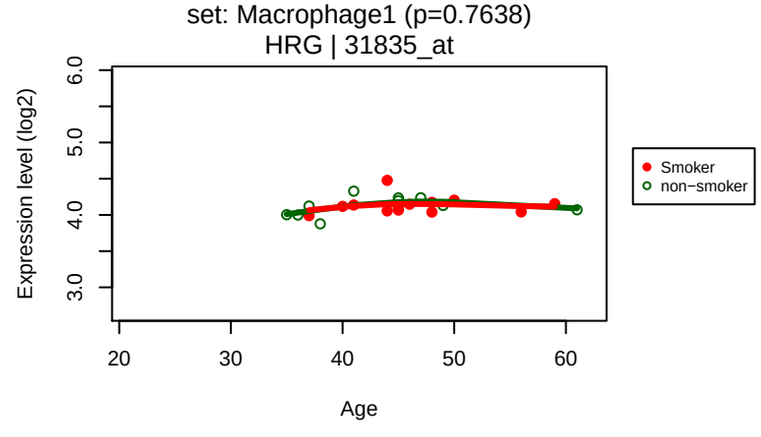

set: SmallAirway1 (p=0.149)
HRG | 31835_at
Expression level (log2)
Age
Smoker
non−smoker

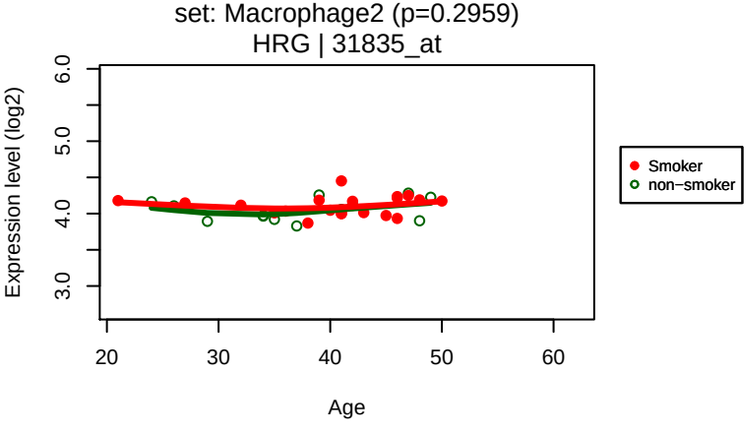

set: SmallAirway2 (p=0.3307)
HRG | 31835_at
Expression level (log2)
Age
Smoker
non−smoker

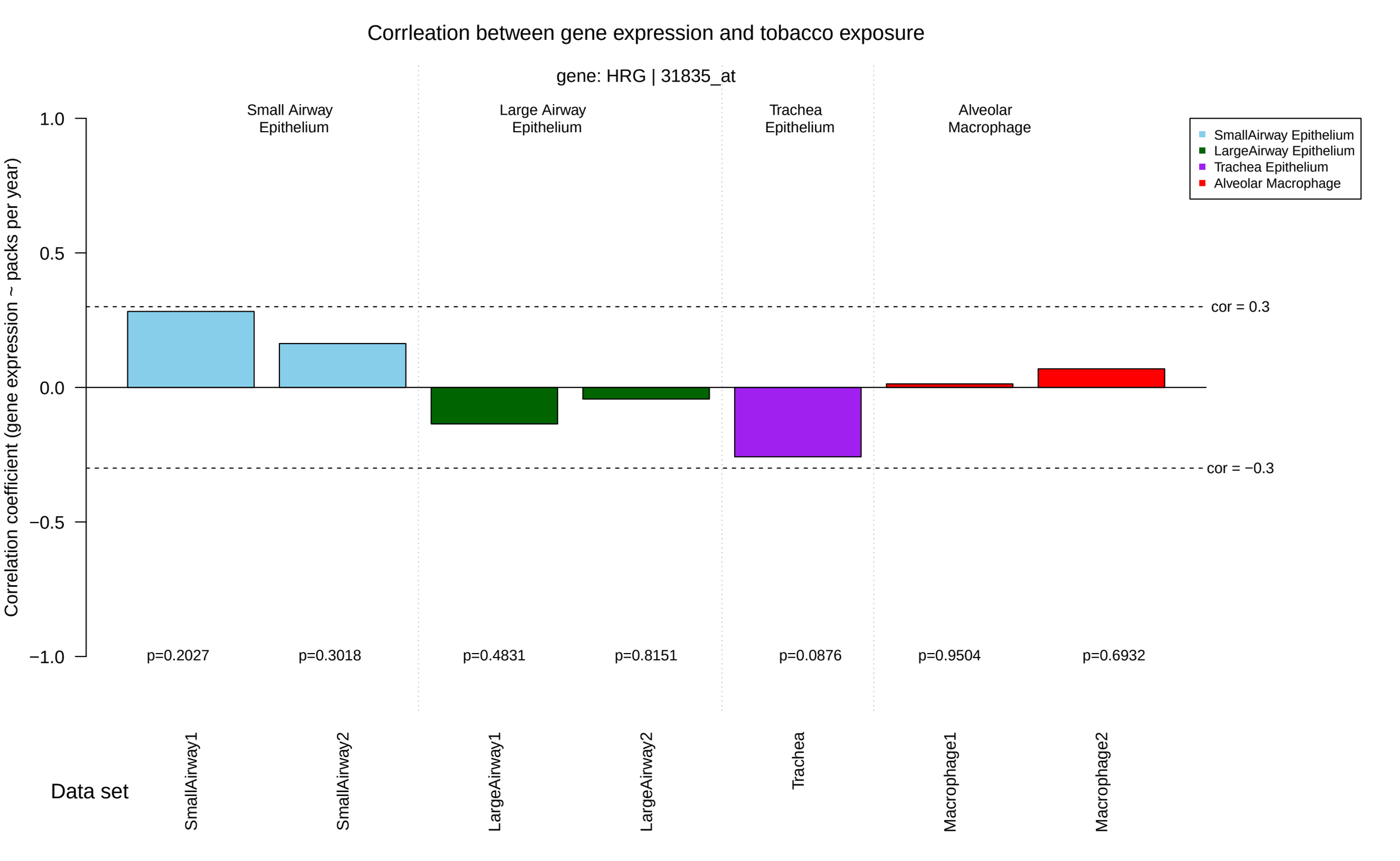

set: LargeAirway1 (p=0.6351)
HRG | 31835_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.508)
HRG | 31835_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4008)
HRG | 31835_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7638)
HRG | 31835_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2959)
HRG | 31835_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HRG | 31835_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2027
p=0.3018
p=0.4831
p=0.8151
p=0.0876
p=0.9504
p=0.6932
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2